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M.Sc. Systems Engineering for Manufacturing

A Structured Literature Review on Supply Chain Mapping

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Non-Technical Project Task Assignment

for Mr. Akshay Dineshbhai CHANDROLA

Topic: A structured literature review on supply chain mapping

Multiple national lockdowns due to the recent COVID-19 pandemic temporarily stopped the global flow of materials. Missing supplies led to disruptions in manufacturing because many companies could not adapt to the new circumstances. These disruptions revealed a significant lack of knowledge that companies have about their supply networks. The knowledge is often limited to first-tier suppliers, whereas higher-tier suppliers are particularly unknown. Supply chain mapping is the process of gathering and visualising information about all entities involved in the supply chain. It is a key measure to enable risk identification and mitigation.

The project work is intended to present the current state of the art in supply chain mapping by using a structured literature review approach. The objective of this work is to identify approaches to reveal hidden entities in the supply chain, present visualisation techniques, and identify software tools to create supply chain maps. Eventually, the work should provide a short critical analysis of the state of the art and give implications for future research directions.

In detail the following things must be done:

- literature research,
- writing a report (at least 30 pages, at most 40 pages, from the introduction to the summary), and
- giving a final presentation of 15 minutes.

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Abstract

Supply chain around world has been affected by multi-national lockdowns resulted by COVID-19. Pandemic has exposed supply chain network's weaknesses, as companies were not able to fulfill the customer requirements due to lack of knowledge regarding their supply chain. Shortage of crucial material supply led to disruption in manufacturing because several industries could not cop-up with new changes. These complications made companies understand importance of supply chain transparency. For many organizations, governments, decision makers, and consumers, supply chains are a primary concern. Supply chain should be mapped to address numerous supply shortcomings. This report discusses essential requirements of supply chain mapping by reviewing literature on, supply chain risk management, Visualisation techniques, and present supply chain mapping software tools. It identifies various approaches to locate hidden risk in supply chain, and ways to visualise supply network as well as software tools available to map supply chain. Advancement of technology has presented many future research opportunities in this area.

Keywords: Supply chain Mapping, Supply Chain, Potential Risk, Visualisation Methods, Supply Map, Supply Chain Mapping Software

Contents

1 Introduction **1**

2 Methodology **3**

3 Related work **5**

3.1 Approaches to reveal hidden entities in supply chain 5

3.1.1 A concept model 6

3.1.2 Analytic hierarchy process (AHP) 7

3.1.3 Analytic network process (ANP) 8

3.1.4 The hazard and operational (HAZOP) 8

3.1.5 A qualitative value-focused process engineering (VFPE) methodology 11

3.1.6 Knowledge-based system approaches 12

3.1.7 Supply Chain Operating Reference (SCOR) 13

3.1.8 Failure Mode and Effect Analysis (FMEA) 14

3.2 Visualisation techniques 15

3.2.1 Visualising Business Data with Generalised Tree Maps 15

3.2.2 Visualisation Tool to Discover Business Process 16

3.2.3 The Event Tunnel Visualisation for Business Process Data 17

3.2.4 Visualisation of Decision Making 18

3.2.5 Interactive Visualisation 18

3.2.6 Visualisation with Risk Management and Real-Time Monitoring . . . 19

3.2.7 Visualisation in Process Management for Daily Operational Activities 21

3.2.8 Supply Chain Simulator 22

3.3 Software tools to create supply chain map 23

4 Conclusion and Future scope **26**

List of Figures

1 Logical link between search terms 3

2 A concept model 7

3 Overview of Value-focused process engineering 11

4 SCOR based processes of retail enterprise 13

5 Tree-map illustrating the distribution of incidents among various organization representatives 15

6 An organization's data is represented at several levels of granularity 16

7 Associated events in the workplace 17

8 Associated events in the workplace 18

9 Flow diagram for interactive visualisation 19

10 Hierarchical design of Riskvis 19

11 Framework of Riskvis 20

12 User interface of Riskvis 20

13 Calendar map view of company's daily operation 21

14 Waste received in ton at each waste transport centre 22

15 A graphical representation. The number by vertex is the WTC's ID 22

16 Supply chain simulator 23

17 Example of supply chain mapping published by VF on Sourcemap 24

List of Tables

1 List of publications on supply chain risk management, visualisation and mapping 4

2 The Identified risk Groups 5

1 Introduction

Supply chain is an essential part of our life in this era of globalisation. COVID-19 has proven to be a major obstacle to modern supply chain. It has been seen throughout pandemic, many products were extinct, and many sellers has to face scarcity of stocks. The main issue observed was a lack of response in organizational supply chains, which shown incompatibility with individual value chains rather than accumulating purchases. However, COVID-19 has seriously affected entire world, including the food supply chain. In addition, automotive industries has faced temporary shut-down as procurement could not fulfill requirement of crucial components [1].

Although the repercussions of the pandemic were severe, supply systems throughout the world did not adapt satisfactorily to them. Information regarding the supply chain must be available in order to implement suitable and successful countermeasures against supply risks. However, studies show that businesses frequently lack the information regarding the higher tier of their value chains. By the beginning of February 2020, more than 60 percent of businesses were still manually determining whether they had indirect suppliers in China, and even global corporations frequently are unaware of their reliance on Asian imports [1]. Almost 50 percents of German logistics companies have insufficient or extremely poor awareness of their supplier's supplier, according to a comparatively new study on the pandemic's effects [2]. Very little portion of textile manufacturers could pinpoint their product's origins, according to a 2019 poll, while The Sustainability Community claims that more than 75 percent of consumer goods producers have little to no knowledge of their suppliers' environmental initiatives [3].

In addition to the difficulties in managing the pandemic's consequences, shortage of supply chain information poses additional risks to businesses. End users value information about the social activities anywhere along value chain and want to know wherever their products are produced frequently[4]. Because customers hold businesses accountable for their supply chains, breaking ethical or environmental standards increases reputational risk [5]. Companies are now under increasing government pressure to operate strictly in accordance with relevant standards, with the Supply Chain Act being considered in Germany serving as a prominent case [6].

Despite the fact that there are powerful determinants towards supply chain information, the gap to industry trend remains. Multiple obstacles contribute to the problem. Businesses primarily consider information about company's supply chain network as confidential and are hesitant of disclosing it to anyone. However, gathering and analysing supply chain information can be difficult as well. Since value chains are getting more complicated and globalized, similarly, gathering the data is not sufficient on its own. Studies typically suggest using a technique called supply chain mapping to gather supply chain information to aid in strategic decision-making [7]. For instance, a former study recommended mapping as a way to get ready for epidemics in the coming years [8]. Supply chain mapping is act of obtaining and visualizing data about all supply chain participants, It is an essential step in the potential

risk identification and prevention process.

Main focus of this report is on supply chain mapping by discussing major three questions. First, It identifies the approaches to reveal potential risk in supply chain. Second, It discusses visualisation methods to map supply chain. Third, It identifies software tools to develop supply chain maps, and concludes by identifying encouraging directions for future work.

2 Methodology

The present study adopts a systematic literature review approach to address the research questions. The study follows five steps:

1. Identification the research Questions
2. Search for relevant studies
3. Selection of studies
4. Charting data
5. Collating, summarizing, and Reporting the results

First, research question are decided to best address task. I used different sources to collect all previous work done by researchers. SCOPUS, Web of science, Sematic scholar and Science direct is were used as meta-database, as there were not a single database providing all research material required. However, there were many restriction in terms of access and language. I incorporated a Google scholar because it offers an access to multiple meta-database at a time relevant research.Later in the review, I also used a forward- and backward search approach for all important publications. The used databases and the search keywords are displayed in Figure 1 together with the logical connection between them.

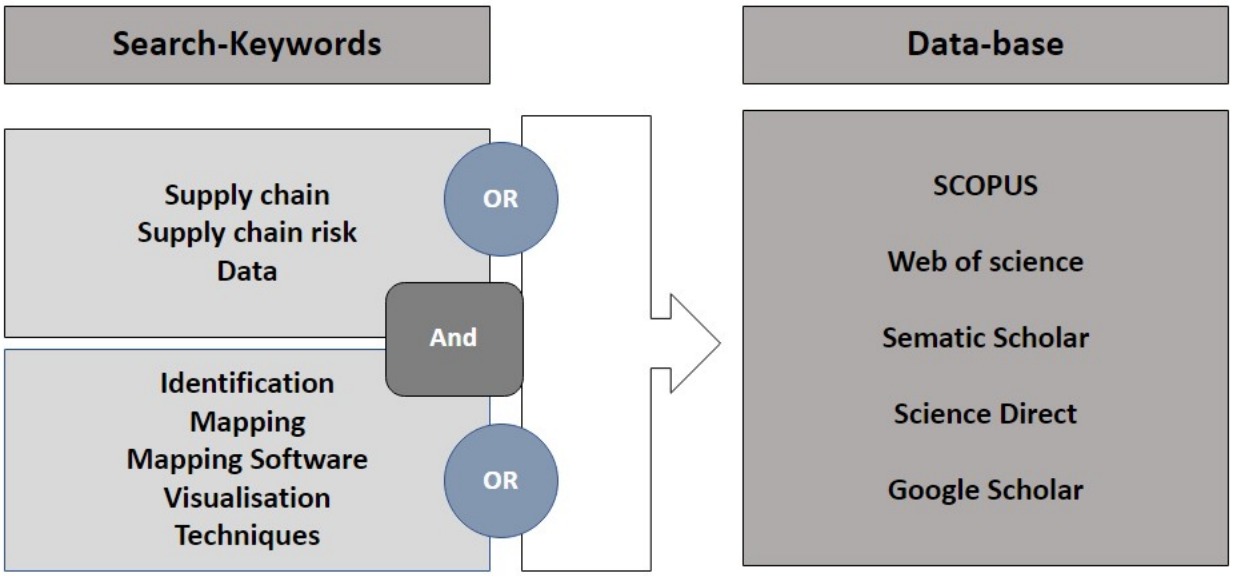

Figure 1: Logical link between search terms

I excluded the term supply chain management from the review because many false-positive results in the search query. Without further delay, I began the initial search for related research on September 10, 2022. Google scholar were used in the first few searches to prioritize

relevancy, afterwards all information was imported in citavi 6 for additional research. By Using the criteria of Title, year, and Authors, I looked for duplicates entries and eliminated them. After eliminating duplicates, I performed a screening process with Title-abstract screening to check relevancy with either supply chain risk identification, visualisation or mapping/maps, and excluded non relevant studies. I also excluded studies in language other than English due to language barrier. I also removed any literature review which did not discuss required fields. Furthermore, I included publication for further analysis. I ended up narrowing down my search to 11 either journal or conference paper, which gave clear information regarding supply chain mapping as well as risk and visualisation refer Table 1

Table 1: List of publications on supply chain risk management, visualisation and mapping

Sr.No.	Title	Reference
1.	International Conference on Business Process management	[9]
2.	Journal of Business logistics	[7]
3.	International Conference Information Visualisation	[10]
4.	International Journal of Production Research	[11]
5.	International Journal of Production Economics	[12]
6.	Journal of Operation Management	[13]
7.	International Journal of Engineering Transaction B.Application	[14]
8.	IEEE Transactions on Visualisation and Computer Graphics	[15]
9.	Supply Chain Management: An International Journal	[16]

During analysing studies from articles in journal and conference paper given in table 1, citation of these articles helped me with further studies search. Section 3 gives detailed explanation regarding findings of this literature review.

3 Related work

This section discusses past research in accordance to supply chain mapping. This will helps the reader to get better understanding of Supply chain maps.

3.1 Approaches to reveal hidden entities in supply chain

Supply chains typically consist of identical agents. However, based on a number of variables, such the activity to which two comparable chains belong, the agents engaging in it, the location where they are formed, or the nutritional content of clients they please, the risks for the two chains may differ. Therefore, it is challenging to presume that a certain risk for one supply chain is equally important and valuable to another supply chain. The risk's greatest flaws are based on how diverse it is, how many different sources it has, how different it's effects might be, and how complex it interacts with other factors. When a risk is identified, it is simply named without being clarified or quantified[17]. The identified risks in supply chain has been categorised in five groups as shown in table 2.

Table 2: The Identified risk Groups

Sr.No.	Group name	Risk
1.	Operational risk	Unanticipated volume peaks Products from the new store are delayed Inventory Breaks for travel and packing
2.	The optimal process for the product or service	An end product of poor quality or Unsatisfactory customer service
3.	Suppliers	Cost of a Good Provider Fraud Default shipping
4.	Safety	Fires Machine failure
5.	Labor rights	Workplace accidents Poor employee training Poor working conditions

Dealing with uncertainty, risk, and vulnerability in supply chains is crucial and challenging, making it a key topic of interest for supply chain management scholars and practitioners in recent years[13]. This section focuses on identifying methods to reveal potential risk in supply chain.

The goal of research is to create methods for identifying, evaluating, analyzing, and treating

supply chain vulnerability and risk. In order to accomplish this goal, research on supply chain risk management tackles issues with the supply chain structure's coherence representation, how risks relate to its elements, the nature of risk, and the key identifying and assessing factors. By ensuring that all "significant activities within the organization have been identified and all the risks resulting from these activities have been defined"[18]. The first set of researchers created methods like, a conceptual models [12], supply chain vulnerability maps[16], and the analytic hierarchy process (AHP) method[19] based on qualitative or quantitative techniques for identifying potential supply chain vulnerabilities. Another team of researchers concentrated on risk factor identification by using AHP[20] and hazard and operational (HAZOP)[21]. Few other supply chain experts proposed qualitative tools to determine both risk categories and risk factors. For instance, a qualitative value-focused process engineering methodology[13] and risk identification system based on knowledge-based system approaches[22].

Furthermore, there are several other identification methods in use to reveal hidden risk in supply chain, such as Supply Chain Operating Reference (SCOR), and Failure Mode and Effect Analysis (FMEA)[14]. These all methods are used to locate possible risk in supply chain, however these methods might need to be combined with each other to achieve best results.

3.1.1 A concept model

Per the Advance Market Research (AMR) research, supplier failure and supply continuity are the real challenges, and acceptance of and spending on Supply Chain Risk Management (SCRM) are rising quickly[23]. It is obvious that in order to control and foresee disruption, it is vital to measure supplier performance. The traditional supplier performance metrics can be divided into strategic (lead time against industry norm, quality, cost savings, and pricing) and tactical (order cycle time, cash flow, quality assurance methodology and capacity flexibility)[24]. Most material supplier evaluation and selection are increasingly considered as strategic issues for businesses. Greater collaboration and reliance on a single source or strategic provider necessitate more sophisticated calculations of the probability of supplier performance glitches[25].

An investigation of possible partners along a variety of criteria is required in order to identify those who fit the existing SC the best. M. Vonderembse [26] created a contingency model of supply chain management (SCM) that examines how supplier characteristics fit with the SC structure and strategy, as shown in the Figure 2 [27] for a proposed approach. The supplier's adaptability and reactivity to market volatility are essential for an agile SC. Reliability and minimal variability in lead times and performance are crucial for a lean SC. Most actual situations demand for hybrid SC's to combine both qualities.

The locations of the suppliers, manufacturing facilities, distributors, and customers in the SC are then split geographically[28]. Large distances can restrict flexibility in the event of severe market turbulence and increase the likelihood of a transportation disruption. The

level of single source has a substantial impact on the risks that a SC faces. Suppliers who serve as a sole source for crucial materials or components deserve special consideration[29].

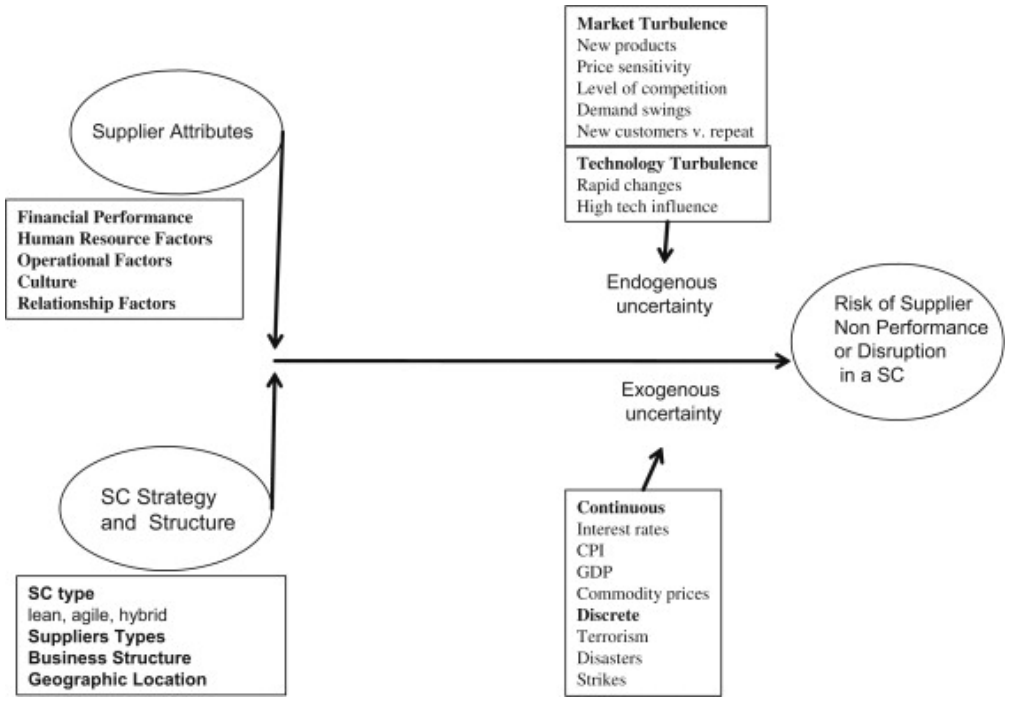

Figure 2: A concept model

3.1.2 Analytic hierarchy process (AHP)

Analytic hierarchy process (AHP) is a mathematical decision-making technique that helps policy-makers to structure decisions, quantify intangible aspects, and evaluate options in a comprehensive and rational framework. Saaty developed the AHP to allow decision makers to reflect the interaction of various factors in difficult circumstances. The procedure involves decision makers to create a hierarchical structure for the explicit components in the given situation, make judgements about the relative importance of each of these factors, and define a preference for each option alternative in relation to each factor [30]. It delivers a prioritized ranking order expressing the general preference for each decision option. AHP's ability to consider both tangible and intangible aspects is a benefit, especially when diverse people's subjective opinions are a key component of the decision-making process[31].

The AHP technique may help managers with a wide range of decisions and complicated issues, such as choosing a supplier, choosing a location for a facility, forecasting, predicting potential risks and opportunities, selecting a technology, designing a strategy and a product, and more [20]. Leading government and commercial companies worldwide, including IBM, Ford, Boeing, British Airways, the Central Intelligence Agency, the Department of Defense, and the Department of State, have successfully implemented the AHP and ANP (The Analytic Network Process, a more sophisticated framework established by Saaty [32]). There

are normally four key types of risk management techniques: Tolerating, Treating, Terminating, and Transferring (known as the 4 T's). Once vulnerabilities have been discovered and assessed, risk evaluation is required to choose the best controlling alternative regarding the treatment of these harmful risk factors [33].

3.1.3 Analytic network process (ANP)

The Analytic Network Process (ANP), which takes into account the interdependence between the hierarchy's constituents, is a generalization of the Analytic Hierarchy Process (AHP). Because they entail the interaction and dependence of higher-level elements of a hierarchy on lower-level ones, many decision issues cannot be organized hierarchically [30]. As a result, a network rather than a hierarchy is used to illustrate ANP.

The feedback structure is not top-to-bottom but rather resembles a network with cycles connecting its constituents, which we can no longer refer to as levels, and loops connecting one element to another. There are sinks and sources as well. A source node never serves as the destination of influential or important pathways; rather, it serves as their origin.A sink node is never the source of channels of influence; rather, it is one of their destinations. Source nodes, intermediate nodes that are on paths from source nodes, lie on cycles, or are on paths to sink nodes, and sink nodes are all possible components of a complete network. Only source and sink nodes may be present in some networks. Others may only consist of source and cycle nodes, sink and cycle nodes, or cycle nodes. In practice, it's common for a choice problem requiring feedback to come up. Any of the aforementioned networks could manifest as it.The difficulty lies in figuring out the relative importance of the network's components, particularly the alternatives to the choice and, more importantly, in defending the legitimacy of the result. The procedures required to obtain the priorities are more complex than in the past since feedback involves cycles, and cycling is an infinite process [30].

3.1.4 The hazard and operational (HAZOP)

HAZOP, which stands for Hazard and Operability Study and is one of the techniques for process safety management, is frequently referred to as a qualitative risk assessment tool, a collaborative approach to risk management, or even a "bottom-up" risk identification technique. A new or current operation method, policy, or standard is reviewed using the HAZOP process .It determines the kind of hazards which might be present in the environment, operator risk and machinery risk. The premise that hazards materialize when deviations from the initial operational or design intent are present is what distinguishes HAZOP from other risk management techniques [34].

The listed hazards are certain to be eliminated when carried out correctly in industries like refining, chemical, oil and gas, pharmaceutical, power generating, and other locations comparable, preventing injuries, fatalities, and equipment damage. HAZOP is a useful

technique that employers can use to increase workplace safety.

This method is designed to look at both the aims and operational variances in addition to the step-by-step procedures. HAZOP works by determining all probable unintentional actions that could occur to reveal potential dangers. Contrary to most conventional hazard identification procedures, HAZOP considers more than just what might happen if everything goes as expected.

Labeling each potential issue with guiding phrases to encourage creative thinking of significant deviations and ensuring their alternatives, is a part of recognizing deviations from a plan or procedure. Since some guide words are more effective at conveying particular circumstances than others, these word sets frequently differ between risk management programs.

When used to describe various nodes or iterations of the plan, each guide word has a predetermined meaning. The following list of standard reference terms is utilized by numerous industries [34]:

- **No or Not** - The complete negation of the intention from the design
- **More** - Everything that is being detected has increased quantitatively
- **Less** - Everything that is being detected has decreased quantitatively
- **As well As** - A change in quality or an improvement in quality has emerged
- **Part of** - There is a change in quality or a decline
- **Other Than** - There was a total replacement
- **Early** - Something happened sooner than expected (clock time)
- **Late** - Something happened later than expected (clock time)
- **Before** - A process step was carried out earlier than it should have been
- **After** - In the process sequence, a step was carried out after it was supposed to

Similar to other hazard management programs, a HAZOP study should be finished in stages as part of a risk management project. A HAZOP assessment should ideally start whenever a new process or a changed process in the workplace occurs. This could be anything from the addition of a new machine to a tweak that makes the work simpler. The strategy largely relies on the following four processes to accomplish this [34]:

Define: The risk assessment team is required to talk about essential concepts at this stage in order to determine where to concentrate their efforts. Define the project's objectives, duties, and team members

Prepare: Guide words are frequently used at this level to assist highlight important concerns in a process, Make a study plan, Gather all relevant information, Select a documentation style, Plan and plan activities and times.

Exam: The chosen guide words are used in the examination step in order to thoroughly look for process irregularities. Additionally, any potential use and misuse scenarios identified by the method must be examined in order to establish their credibility.

- Using guiding words, separate the system in question and look for any probable locations where it differs from design
- Indicate the effects and the causes of certain known problems
- Determine the importance of such issues
- Assemble measures or methods that can identify issues
- Determine protective measures

Document: For this kind of documentation, templates are frequently utilized. The IEC 61882 Standard has one such example.

- Record the inspection and certify the documentation.
- Create a report detailing the risk management strategies
- Observe the adjustments made and, if necessary re-evaluate them
- Create the final report.

This approach needs a team of professionals in areas like operations, maintenance, instrumentation, engineering, process design, etc. to be functional. Depending on the process or technique being reviewed, the appropriate team size will vary, but it should be maintained as small as possible without compromising the necessary ability or knowledge [34].

When it comes to hazard analysis, HAZOP is a crucial instrument. In reality, it has a number of advantages that are specific to itself. The first is its effectiveness for dealing with threats that are challenging to measure, such as those that are based on human behaviours and are challenging to identify. The reason is that HAZOP doesn't specifically request that the risk management team measure this kind of departure from the usual. Instead, they are required to employ guide phrases that enable them to foresee dangers before they arise. HAZOP also includes a brainstorming tool to aid with hazard prediction. It is strongly reliant on this element. Finally, HAZOP is a systematic and complete procedure which is considerably easier to apply than other traditional risk assessment methodologies [34].

In terms of drawbacks, HAZOP does not have the capability of assessing dangers associated with the interplay of various processes inside systems. This method also lacks a prioritization strategy. Finally, there is no means for users to evaluate the effectiveness of new or existing controls.

3.1.5 A qualitative value-focused process engineering (VFPE) methodology

Value-focused process engineering (VFPE) is a methodology that integrates process-based and objectives-based business modelling techniques into a model that allows for comprehensive business representation. VFPE establishes operational and strategic relationships between business processes and company objectives [9, 35, 36] by incorporating value focused thinking (VFT) [37] and extended event driven process chain [38].

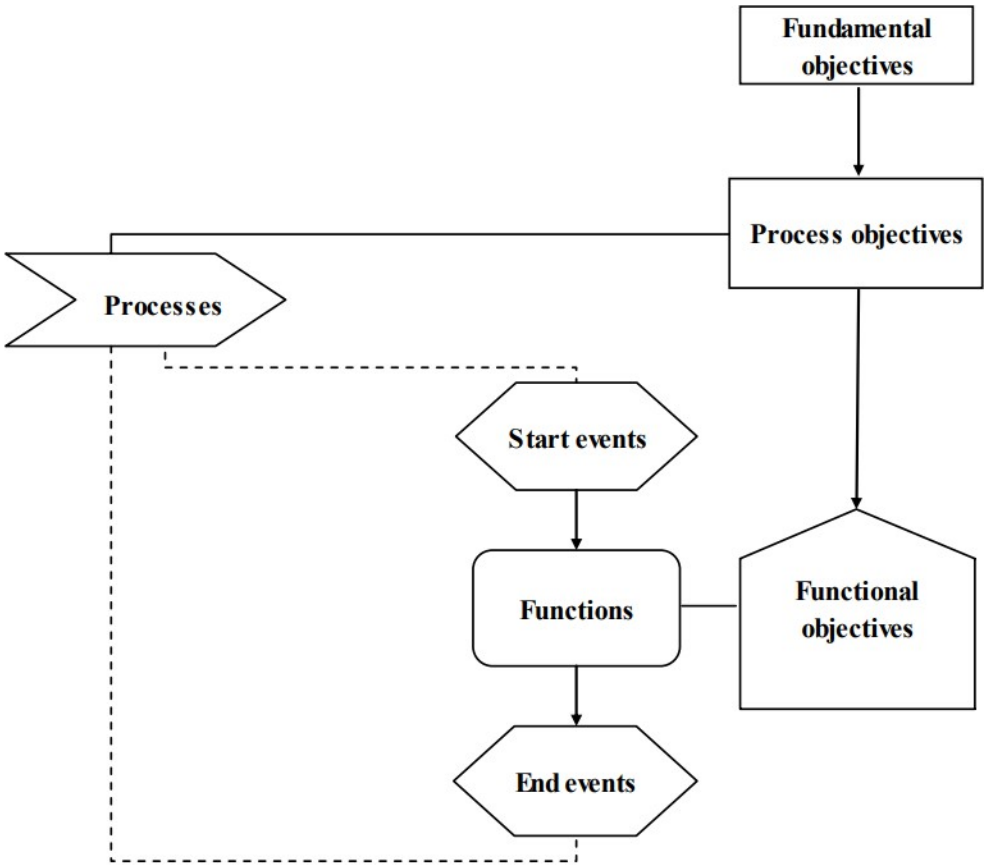

Figure 3: Overview of Value-focused process engineering

VFT differentiates between two different types of objectives: basic and mean. The values of an organization are expressed in the concept of primary objective, which is specified by Keeney [37] "as a statement of something that one wants to strive toward". Fundamental goals are organized in a rigid objectives,a hierarchy that connects lower and higher level goals. Mean aims are defined as "methods to achieve ends" by Keeney [37] at the operational level. These goals are organized into a network of goals. Mean objectives in the VFT serve as a link

between conceptual high-level corporate ideals and quantifiable decision alternatives,which connect to business activities and procedures.

As illustrated in figure 3 [27], business processes and objectives are identified and synchronized concurrently in order to build a clear relationship between the process and objective structures in the context of value-focused process engineering.

3.1.6 Knowledge-based system approaches

Expert systems (ES) or KBS are a part of artificial intelligence (AI). It is described as a computer program that combines a knowledge base and an inference engine to offer features for problem-solving that are intended to replicate the abilities of an expert [39]. KBS is made up of various parts, the most important of which are as follows [40, 41]:

- The knowledge base, which makes up the majority of KBS, is made up of factual knowledge (facts) and rules (heuristics). Knowledge may be modeled using a variety of methods, including rule-based (production systems), fuzzy logic, induction, case-based reasoning, neural networks, and object-oriented systems.
- By using the proper rules, the inference engine generates a conclusion or proposal based on the most recent information in the knowledge base. There are two methods of reasoning that are frequently employed. Both forward and backward reasoning/chaining are used. Backward chaining is a goal-driven approach because it establishes the existence of the "if" part before concluding the "then" part, whereas forward chaining is a data-driven approach because it concludes the "then" part when the "if" part is given.
- In addition to storing new facts by incorporating updated data produced by the inference engine, working memory also stores problem-specific information from user consultation sessions.
- KBS and the user can communicate (dialog) through the user interface.

For the development of a KBS/ES, three different types of computer programming are available: conventional language, AI language, and ES shell. The information base offered by ES shell is empty. In contrast to conventional and AI language, it also features built-in inference capabilities and a user interface. Moreover, the development of KBS utilizing both conventional and AI language requires highly skilled programming. Because of this, ES shell is a more widely used programming language [42, 43].

KBS has many benefits, including faster decision-making, retaining and storing tacit information from specialists and experienced personnel, and facilitating knowledge transfer [34]. KBS has been used in numerous industries, including banking, manufacturing, management, construction, biotechnology, and the military, as a result of these tempting advantages [44]. Piramuthu [45] used the KBS approach in the SCM environment to create an automated

supply chain configurer (ASCC). According to the requirements of the customer order, ASCC recommends the appropriate SC configuration. Next, Chan et al. [46] proposed an experience and understanding simulation platform (KBSP) that may help decision makers and trainees engage with their suppliers and retailers while implementing a vendor managed inventory (VMI) approach. This system might forecast performance by simulating SC (suppliers, manufacturers, and retailers) operations. A further application of KBS is the creation of supplier selection in the automobile sectors based on Yigin [11].

3.1.7 Supply Chain Operating Reference (SCOR)

SCOR is a hierarchical cross-functional process reference model. The process that defines

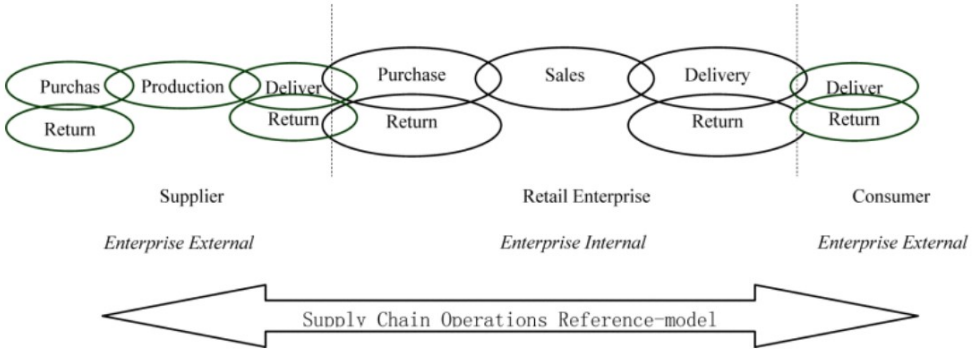

Figure 4: SCOR based processes of retail enterprise

SCOR may be broken down into three layers, and each layer can be analysed to examine, how the supply chain functions. The five fundamental processes are described in the first layer of the SCOR: plan, source, make, deliver, and return . It establishes the parameters of the supply chain operations reference model and the framework for setting competitive performance goals. Companies may make fundamental strategic decisions based on supply chain performance by examining the first layer of SCOR. The retail enterprise supply chain risk management procedure based on SCOR is illustrated in Figure 4 [47].

The SCOR model-based process for identifying supply chain risks primarily uses the network of connected components in the supply chain to identify all relevant risk factors and uncertainties. The identification of risk factors is the first step in the fundamental supply chain risk management methods. It also serves as the foundation and challenges of supply chain risk assessments.

3.1.8 Failure Mode and Effect Analysis (FMEA)

Failure Modes and Effects Analysis (FMEA) is a method for systematically finding probable flaws in a product or process. Consider this: during an FMEA, you will examine materials, assembly, sub-assemblies, and production lines to identify potential failure modes and take steps to eliminate them [48]. In these situations, common risk-quantification variables include:

- How frequently a defect-related activity occurs
- The number of defective parts
- The capacity to identify the problem
- Probability of the defect
- The defect's severity

For every prospective failure, a risk priority number (RPN) is derived, with a common RPN being the sum of the probability, tractability, and severity of the failure. For the majority of businesses, FMEA is a common tool that is used to gather data for risk management choices in an engineering capacity but not in a supply chain functions. The majority of managers approved a modified version of the instrument that could be used to assist determine the risk of SCM operations. Following are the seven steps needed to complete an FMEA [49]:

1. Determine risk classifications.
2. Recognize possible dangers
3. Assess each risk's opportunity, likelihood, and seriousness.
4. Determine the RPN for every risk.
5. Utilize tools like a Pareto distribution to analyze risks using RPN.
6. Create countermeasures to hazards with a high RPN.
7. Re-evaluate risks using a new FMEA cycle.

The majority of professionals agreed that, proactive risk mitigation activities applied to the supply chain are necessary for minimizing disruptions. There was a common perception that unforeseen issues that could have influenced the success of SCM activities could be minimized with an FMEA-based SCM risk assessment tool [49].

3.2 Visualisation techniques

The supply chain is critical to the development and operations of any organization. Regardless of size, a well-organized supply chain helps the company and ensures the continuation of its operations and future growth. An ineffective supply chain undermines the company and may result in significant hidden costs that are bad for business and impede its expansion. An organization works to develop dependable and effective communication channels for better supply chain knowledge and maintenance. Visualization in the supply chain not only offers chances for effective communication but also makes it easier to comprehend and make decisions. It also helps with inefficient supply management of activities.

3.2.1 Visualising Business Data with Generalised Tree Maps

Businesses gather data to assist in decision-making and to provide information. Visual representations of the data, such as bar charts and pie charts, are the conventional method for presenting information. However, because they are less well-known and efficient than business graphics, tree-maps are rarely employed to depict hierarchical data. For the purpose of addressing the creation of useful tools for working with tree-maps in supply chain data, Vliegen et al [15]. introduced a generalized tree-map method. With the help of the suggested improvements to the tree-map method, supply chain data can be represented in a manner that is simple to understand for business users and that also resembles business visualizations [10]. The proposed tree-maps integrate corporate graphics' strong characteristics, such as ease of use, understanding, and detailed level of information availability. The suggested approach includes various layout algorithms that can be used at numerous levels to maximize benefits. This type of presentation enables for identification and the extraction of relevant insights while leaving the scope of similar work for mapping on geographical maps. Figure 5 [50] is a sample tree map produced by the proposed method.

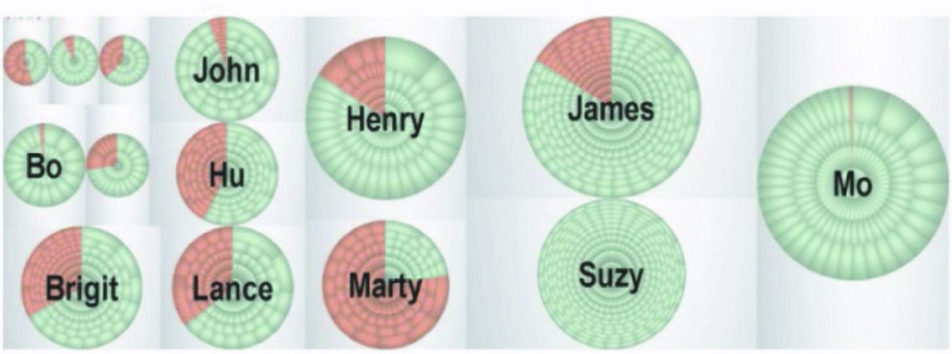

Figure 5: Tree-map illustrating the distribution of incidents among various organization representatives

3.2.2 Visualisation Tool to Discover Business Process

It is common practice in business decision making to differentiate management of company operations from management of business processes. However, such separation causes an information vacuum and emphasizes the importance of communication across different elements of the business. User reports and dash boarding are commonly used to automate reporting and the creation of business insights from supply chain data. Such an approach places constraints on the usability and effectiveness of presenting data insights via relevant data, viz. The ultimate implementation of such systems is complicated by the varying levels of information granularity demanded. Urabe et al. [51] present an overview of a visualisation tool comparable to Disco [52] and Celonis [53], which provides an example of an effective solution to such a challenge. It has been established that making information available at various levels of granularity with distinct visibility layers improves the ease of use and efficiency of decision making. The tool displays information at the desired level of information and provides a major advantage for large organization's supply chains to leverage high information volume and a greater number of information flows and levels [10]. Figure 6 [51] shows how the tool is used with this type of corporate entity and displays the appropriate information flows and levels.

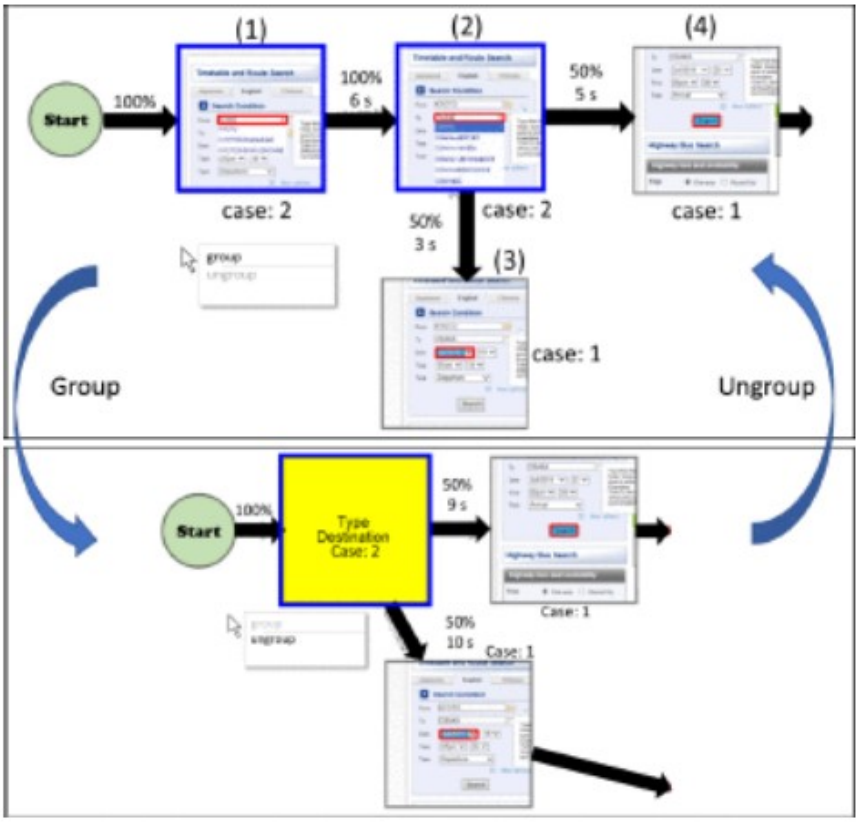

Figure 6: An organization's data is represented at several levels of granularity

3.2.3 The Event Tunnel Visualisation for Business Process Data

In a modern world, business processes are increasingly becoming event-driven and event-based loosely linked distributed systems, which means that the event state changes over time and a retracing method is required to discover any abnormalities or errors [54]. Figure 7 [54] depicts an event tunnel that stores information about business activity. The event data is stored in a 3D cylinder, which provides the user with two visualisations. The first is a top view that shows events along the time axis, and the second is a side view that shows the chronological order of the events. The top-view event tunnel's 3D view is projected to circular 2D rendering. In top-view visualisation, the most recent developments and work processes are at the top of the tunnel. These occurrences are also more relevant for the business analyst because they occurred recently. It also helps in displaying connected occurrences that are near to one another.

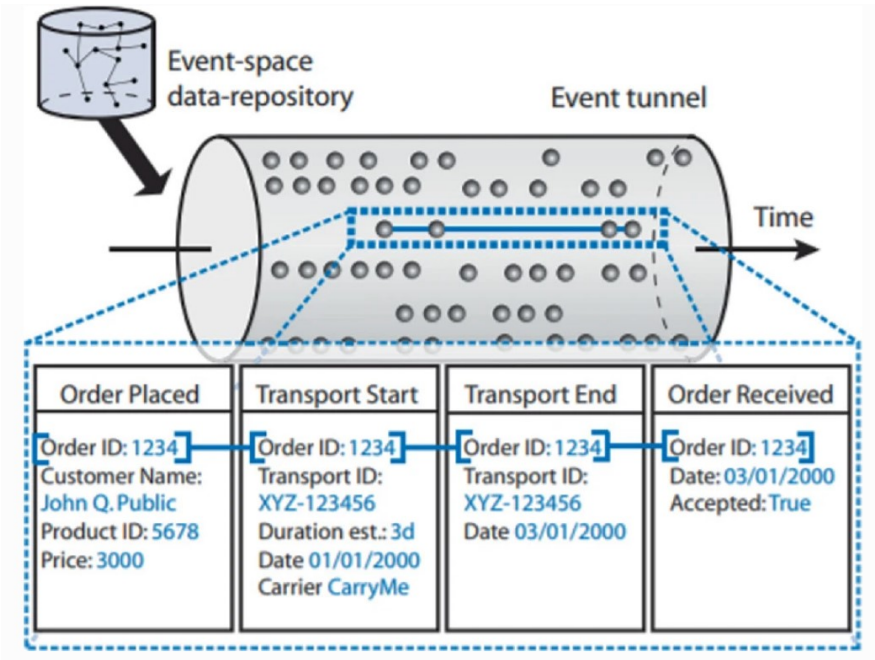

Figure 7: Associated events in the workplace

The second visualisation is the side view, which tells the order of the event and correlates event sequence. The visualization is similar to a process chart, such as a GANTT diagram, in which the schedule of each event is given. Figure 8 [54] depicts a dashboard with (a) the event tunnel side view, (b) a text view, and other (c,d,e,f,g,h,i,j) configuration consoles. This type of visualization is appropriate for big amounts of data in a business process with up to 40,000 events. The event tunnel visualization can be used in real-time resource and logistics management. This has two benefits: an event group can be reduced to a single visual sequence pattern, and correlation between several business processes can be obtained.

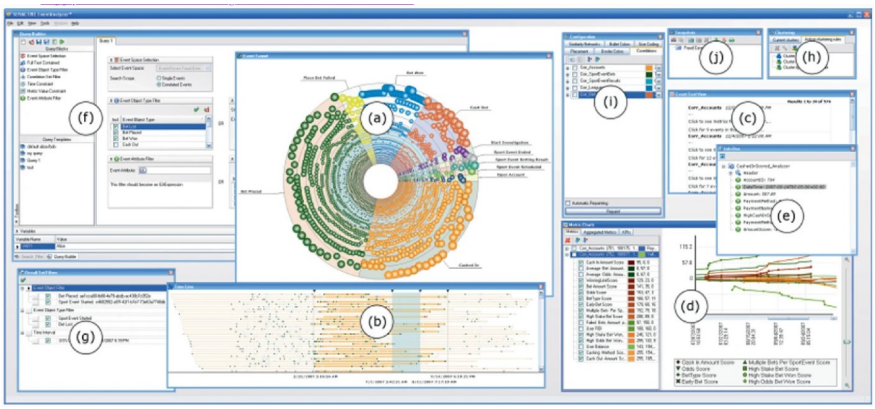

Figure 8: Associated events in the workplace

3.2.4 Visualisation of Decision Making

The visualisation of decision making is the second big subject discovered by the content analysis. This topic merits special attention as a crucial stage in logistics and supply chain management because it draws on a broad range of data visualisation and information system management literature. The visualization of decision making is the second main subject highlighted by the content analysis. This issue merits special attention as a crucial step in logistics and supply chain management because it pulls on a wide range of data visualization and information system management literature. Organizational management operations are heavily reliant on the availability of data and the creation of data insights. The visualization of organizational data opens up new possibilities in all aspects of logistics and supply chain management. The separation of control and operations creates substantial difficulties for decision makers who must lean on data insights for their validity, credibility, and agility. The ability to respond quickly to rapidly changing supply chain conditions places an organization at risk. In this context, visualization Visualisation is a method of presenting a big volume of data in a quick and diverse fashion, allowing information to be delivered to the user more efficiently, resulting in optimal strategic planning.

3.2.5 Interactive Visualisation

Interactive visualisation is one of the themes classified under the broad subject of decision-making visualization. Singh and his team [55] are an example of a paper in this field that is in great demand in the industry because of the inherent advantages of interactivity. They proposed a web-based interactive visualisation prototype for displaying supply chain data. The prototype, built in the Django framework, aims to help managers make decisions by presenting alternative data views of the same supply chain. This prototype provides the user with predictive analytic opportunities by combining data visualization with analytical skills to provide relevant insights. Each graphic layout sheds light on a different part of the supply chain network. The prototype provides access to visual layouts from a specific point and guarantees that no critical information for decision making is lost. Figure 9 [55] shows

the suggested prototype's flow diagram.

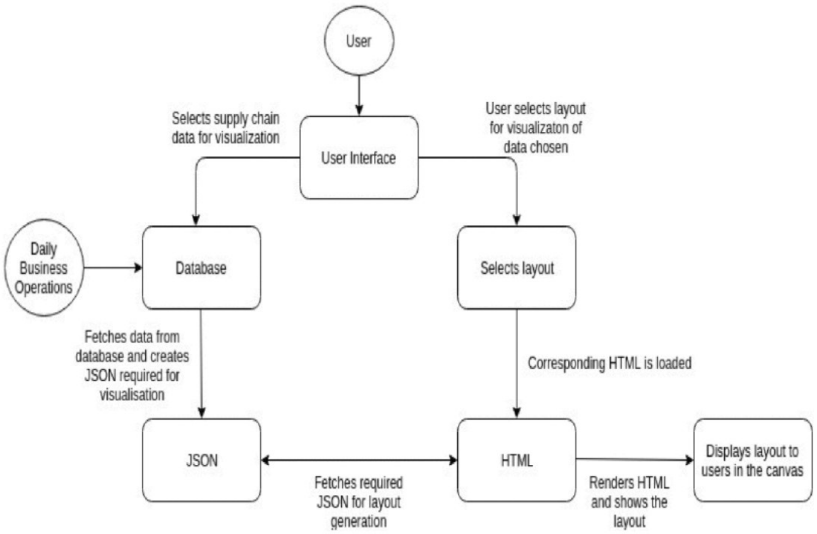

Figure 9: Flow diagram for interactive visualisation

3.2.6 Visualisation with Risk Management and Real-Time Monitoring

As the supply chain becomes more complicated, visibility of the global supply chain has become an issue to address. Goh et al. [56], for example, introduced RiskVis, a supply chain visualisation platform that uses a multi-hierarchical module designed to address complicated supply chain network visibility concerns.The tool's major goal is to manage risk and monitor it in real time for more efficient and effective supply chain risk management. The proposed

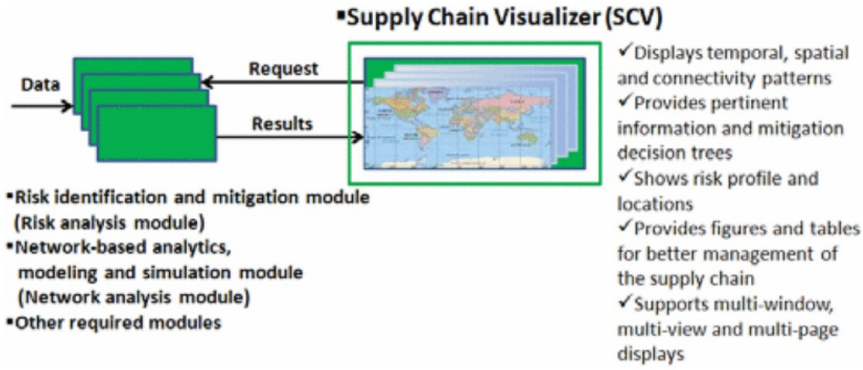

Figure 10: Hierarchical design of Riskvis

supply chain visualiser tool provides management with a clear view and data

insights into supply chain operations in local as well as global level.

This platform discovers Spatio-temporal connection patterns of supply chain elements. Furthermore, using a hierarchical technique, the platform can detect real-time hazards associated with data gathering, risk monitoring, and data insights creation. The suggested visualisation approach enables effective study of massive data-sets, including statistics on a target firm and probable supplier substitutes. Another use of visualisation in the platform is real-time anomaly identification and the fast deployment of personnel to handle abnormalities. Figures 10 [56], 11 [56] and 12 [56] show an example of how the platform might be used for supply chain visualisations.

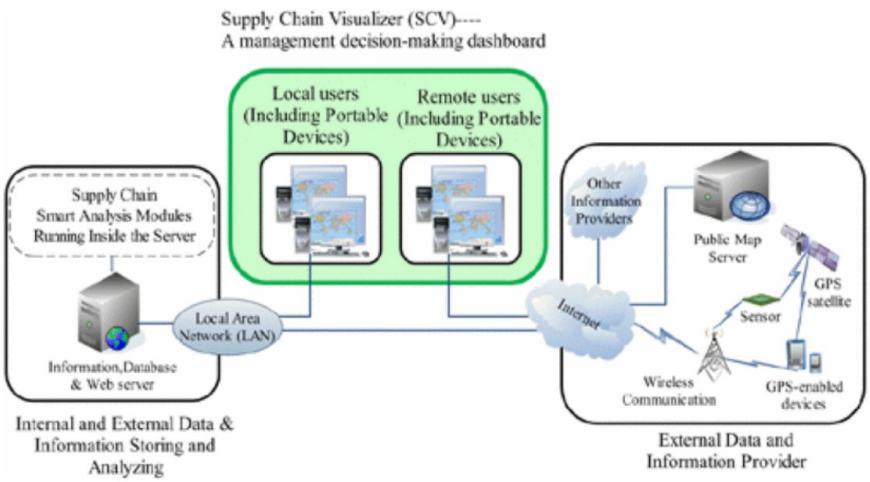

Figure 11: Framework of Riskvis

Figure 12: User interface of Riskvis

3.2.7 Visualisation in Process Management for Daily Operational Activities

Supply chain decision making is closely integrated with the production process, allowing the merger of new manufacturing technologies. Just in Time Lean Manufacturing is one such latest development. Just in Time Lean Manufacturing, which is widely used in the manufacturing industry, satisfies the needs of organizations by allowing cost reduction and increasing operational efficiency. However, the mentioned advantages come at a higher cost with these variants. For example, any outliers in production tend to line up and might lead to significant expenses later on. To remedy this issue, several articles have been offered, namely [57]. They developed a system concentrated on the process and presentation of manufacturing sequential data in order to rapidly convey the production status to management, supervision, and operational staff. In manufacturing process management, appropriate lo-

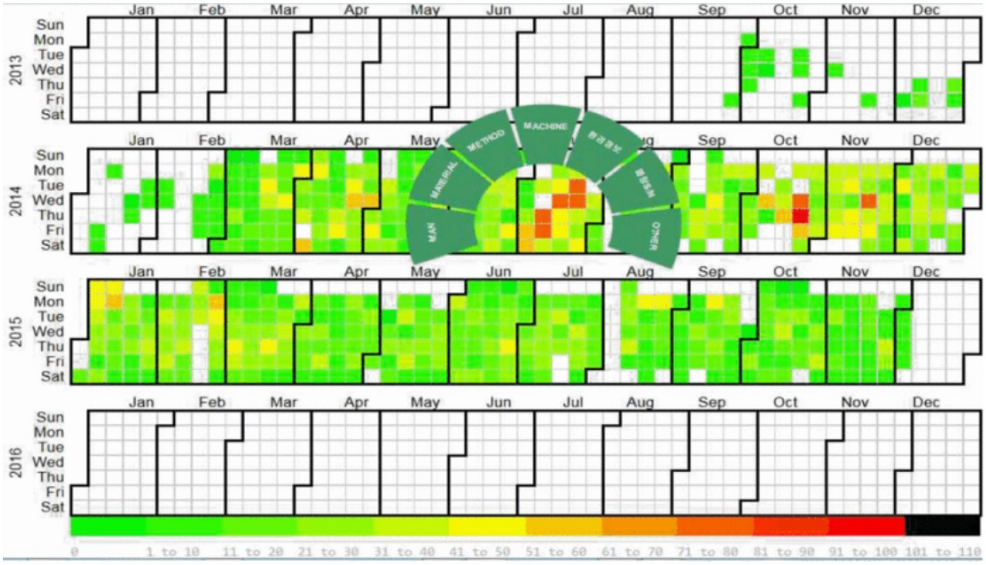

Figure 13: Calendar map view of company's daily operation

gistics on everyday tasks is also crucial for the firm to prevent costly expenditures, excessive costs, outages, and save time. Though a comprehensive assembly line plan is sufficient to get an assembly line up and running, any visualisation of the maintenance and status of the same would provide the greatest results in keeping the best manufacturing process outcomes. Developed the concept of visualizing production data in a calendar view map, see figure 13 [58]. This suggested approach is capable of exposing any risk factors in the production line, allowing the company to act swiftly and restore the ideal production line.

Geographic Information System (GIS) is used in supply chain planning approaches to collect real-time data from the road , allowing operators to acquire a detailed picture of the application. As shown in part in Figures 14 [59] and 15 [59], the graphical method suggested by Lou et al. [59] has also supported the best transportation strategy for residential waste that is identified.

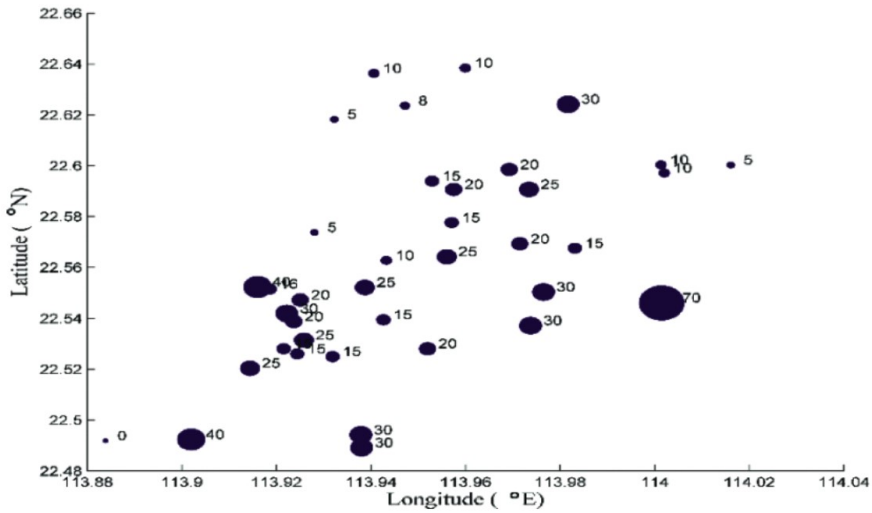

Figure 14: Waste received in ton at each waste transport centre

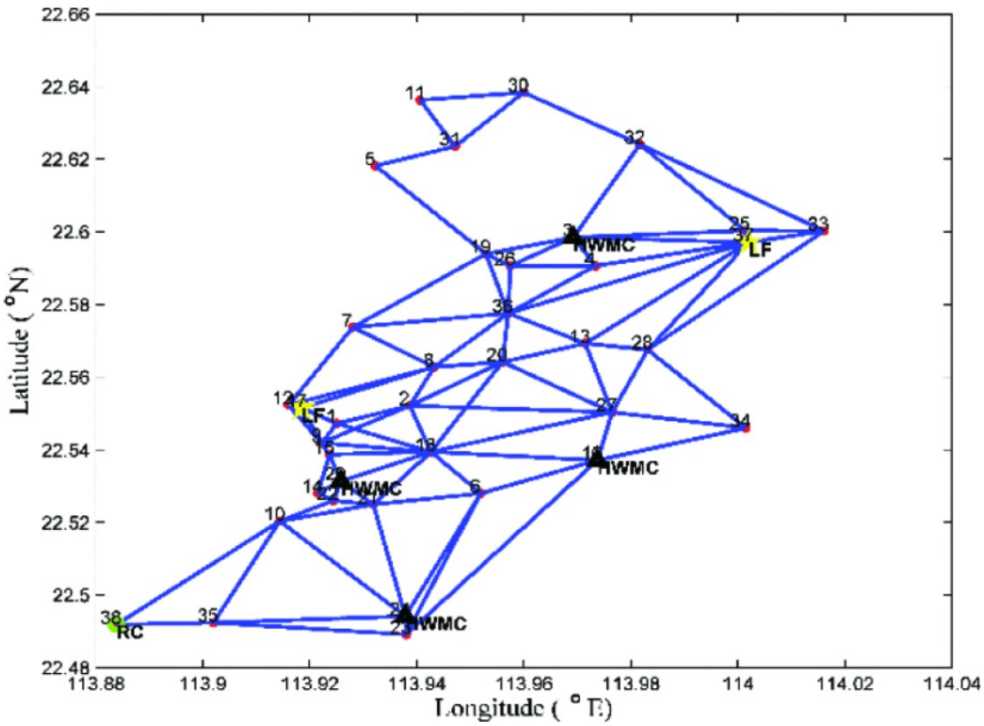

Figure 15: A graphical representation. The number by vertex is the WTC's ID

3.2.8 Supply Chain Simulator

Supply chain simulations are another way that visualization techniques are being used for decision-making. The suggested simulator, which gives a supply chain simulation model, includes extensive information for the user to help with decision-making as well as the representation of the actual behaviour of the complex system [60]. The supply chain network

of factories and distribution networks that executes the process of acquiring and transforming materials from producer to consumer is represented by the model. The interface of the suggested model that provides access to the supply chain simulator is shown in Figure 16 [60].

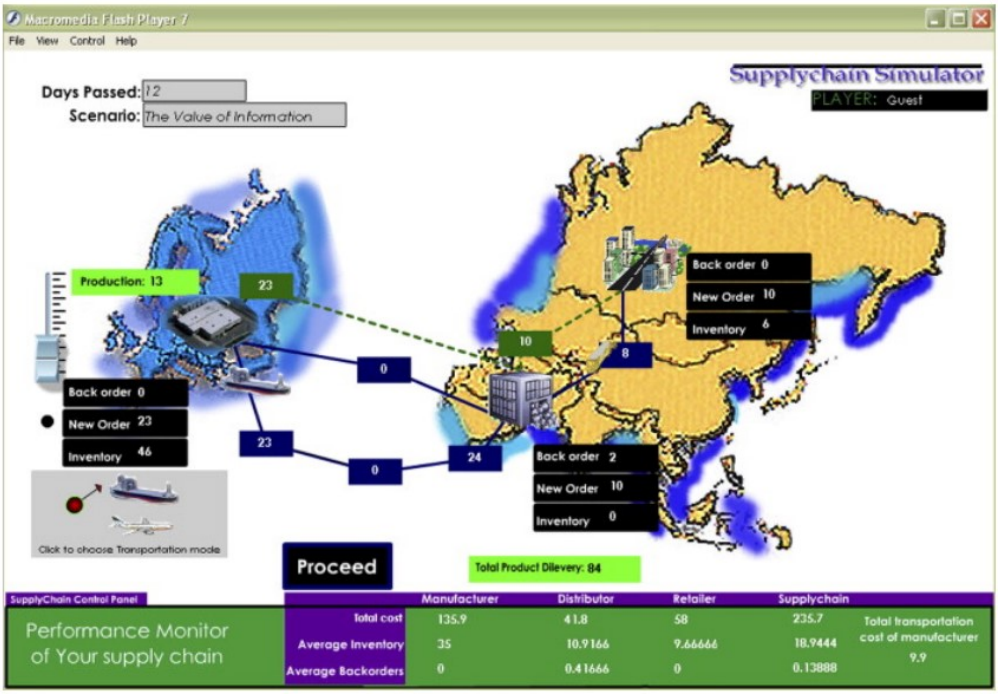

Figure 16: Supply chain simulator

A decision-maker in an administrative position must carefully consider how information is presented. Management needs timely access to business information to take decisions quickly and effectively, especially in today's business environment, which is controlled by massive and complicated supply chain networks.

3.3 Software tools to create supply chain map

Over the past ten years, a number of commercial solutions for supply chain mapping and end-to-end traceability have been developed. These solutions incorporate additional private and public data and information networks with the data sources and techniques described above. In response to the current emphasis on supply chain visibility, traceability, flexibility, resilience, and sustainability, a number of suppliers offer IT solutions and consulting services. Digital technology has accelerated the automated collection and updating of data for supply chain maps in this fast-evolving sector. Only a few examples of such software tools are noted in this report. Report does not aim to be in depth or to assess the capabilities of individual solutions.

A number of all-purpose enterprise supply chain planning tools, including Blue Yonder [61],

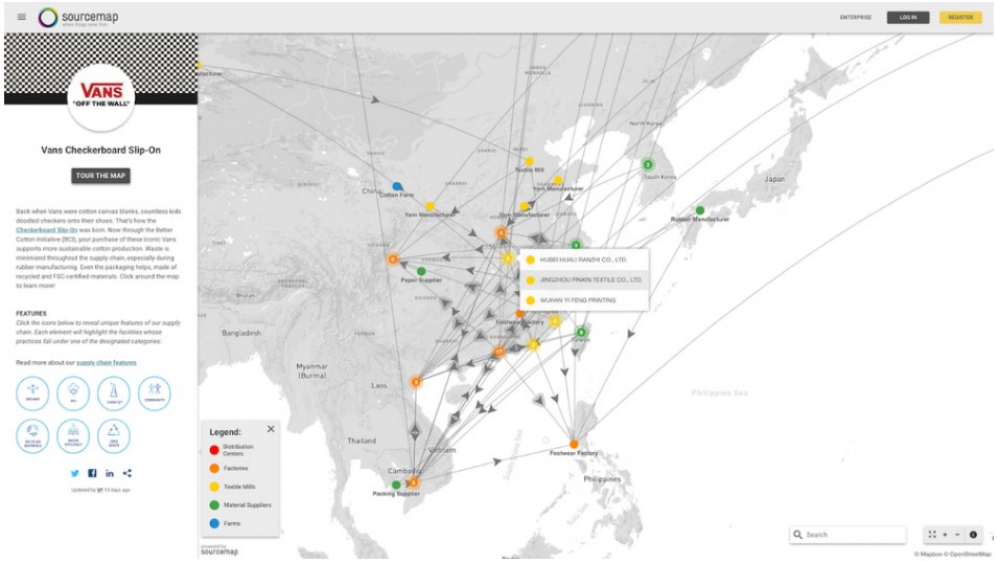

Figure 17: Example of supply chain mapping published by VF on Sourcemap

SupplyOn [62], Elementum [63], Coupa [64], Sourcemap [65] and E2open [66], offer supplier visibility and integration solutions by connecting the planning systems of the center firm to data on the effectiveness and development of suppliers [67], see figure 17 [65] In order to integrate first-tier suppliers to the platform, the mapping process often begins with the delivering of invitations and surveys to them. The process of inviting new supplier is coordinated up to the level of raw materials to map the higher tiers. To achieve this, the key company must have the authority to ensure that both its direct and indirect suppliers sign up for the platform. Data on material flow, a crucial component of supply chain maps, and supplier risk evaluations utilizing third-party financial, geopolitical, climatic, and slavery risk databases provide additional support for organizational transparency. The Open Apparel Registry [68] effort standardizes and exchanges data on international supply chains at the company and facility level given by owners, vendors, Nongovernmental organizations, and certification schemes in the apparel sector with a focus on transparency and collaboration [67].

Moreover, a large number of digital technological start-ups have created alternative approaches for mapping supply chain that employ Big Data. By combining numerous data sources on export, merchandise deliveries, sales invoice, and company relationships with analytical techniques, such as capturing facility locations and identifying goods that are being dispatched, Everstream Analytics [69], formerly DHL Resilience360, and Altana AI [70] aim to map multi-tier supply networks. In a similar manner, Makersite [71] offers multi-tier transparency solutions by combining a variety of data sources with Product Lifecycle Management and sourcing information. FRDM [72] employs artificial intelligence to forecast the Bill-of-Materials framework for acquired components with an equal sustainability focus. Similarly, OpenSC [73] tracks specific items using different technologies like RFID, QR codes, and blockchain and then validates ethical and ecological supply credentials at the source. This information is then communicated with stakeholders, including customers. Companies like BlocRice [74] are experimenting with blockchain technology to track items. Nevertheless,

such real-time validation, mapping, and monitoring involves an expense that may be difficult to establish and call for the cooperation of several stakeholders. Additionally, in order to be able to recognize and include various supply chain entities, supply chain network mapping is often required. As an alternative, VersedAI [75] use Natural Language Processing methods to retrieve buyer-supplier relations from public data. There is a growing industry and a wide range of different techniques and information sources used, as it is clear from the large number of companies providing supply chain mapping and transparency software solutions [67].

4 Conclusion and Future scope

COVID-19 has affected supply chain around the world. This report has identified ways to find out supply chain risk and visualisation techniques to prevent any disruption which might take place in future. Supply chain mapping benefits organisations to track and prevent any risk in their supply chain by increasing transparency. However, there are not enough data available to map each tier supplier in supply chain, and every individual might not provide data for supply chain mapping, this is huge limitation to visualise entire supply chain.

Despite the numerous difficulties, the modern world is becoming more and more dependent on automation technology. New software tools development has the potential of making supply chain process quicker and transparent. Although, they will not be immediate solution, the researcher has high hopes for exciting results. There are plethora of research scope, this report focuses on three. The first area of focus is the development of maps that are focused on the sustainability of supply chains and the shortcomings that can be identified through various computational network analysis [76]. The second involves combining supply chain mapping with geographic information systems, a research field which has theories and practices, that might enhance supply chain mapping but is still in its development phase. The third is the application of mapping to support the digitisation of supply chain network, particularly when integrating with cutting-edge digital technology [77].

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